COMPLETION ALONG A CLOSED SUBSPACE INDUCES A BIJECTION ON MORPHISMS INTO A MODIFICATION WHICH IS AN ISOMORPHISM OVER THE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an injectivity step, an etale covering gluing step, and a two-stage reduction to the affine case closed by the rig-etale algebraization lemma, all carried out inside the delivered span

Editorial note. The label lemma-completion-functor-fully-faithful names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly crashed during packaging (stale empty job directory retained as evidence).

Allowed standard prerequisites (editorial).

- The completion construction Y maps to Y_{/T} for algebraic spaces over the pair (X, T), as set up in the surrounding section of the source
- The fppf sheaf property of the algebraic space X' and descent of morphisms along etale coverings
- Formal algebraic spaces: the reduction of a completion is the reduced induced scheme on the closed subspace

Source reference. Original equation 0880: sources/stacks-restricted/pinned-original-source.tex, source lines 6766–6766. Prerequisite pointer invoked at proof line(s) 6840; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GB9: sources/stacks-chow/tags.txt, source lines 20364–20364. Prerequisite pointer invoked at proof line(s) 6896; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GB8: sources/stacks-chow/tags.txt, source lines 20363–20363. Prerequisite pointer invoked at proof line(s) 6914; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR3: sources/stacks-restricted/pinned-original-source.tex, source lines 6425–6425. Prerequisite pointer invoked at proof line(s) 6856; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR4: sources/stacks-restricted/pinned-original-source.tex, source lines 6406–6406. Prerequisite pointer invoked at proof line(s) 6915; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0240: sources/stacks-restricted/pinned-original-source.tex, source lines 6502–6502. Prerequisite pointer invoked at proof line(s) 6848; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above. Let $f : X' \to X$ be a morphism of algebraic spaces which is locally of finite type and an isomorphism over U. Let $g : Y \to X$ be a morphism with Y locally Noetherian. Then completion defines a bijection*

$$\mathrm{Mor}_X(Y, X') \longrightarrow \mathrm{Mor}_{X_{/T}}(Y_{/T}, X'_{/T})$$

In particular, the functor (0880) is fully faithful.

Proof. Let $a, b : Y \to X'$ be morphisms over X such that $a_{/T} = b_{/T}$. Then we see that a and b agree over the open subspace $g^{-1}U$ and after completion along $g^{-1}T$. Hence $a = b$ by Lemma 0240. In other words, the completion map is always injective.

Let $\alpha : Y_{/T} \to X'_{/T}$ be a morphism of formal algebraic spaces over $X_{/T}$. We have to prove there exists a morphism $a : Y \to X'$ over X such that $\alpha = a_{/T}$. The proof proceeds by a standard but cumbersome reduction to the affine case and then applying Lemma 0AR3.

Let $\{h_i : Y_i \to Y\}$ be an étale covering of algebraic spaces. If we can find for each i a morphism $a_i : Y_i \to X'$ over X whose completion $(a_i)_{/T} : (Y_i)_{/T} \to X'_{/T}$ is equal to $\alpha \circ (h_i)_{/T}$, then we get a morphism $a : Y \to X'$ with $\alpha = a_{/T}$. Namely, we first observe that $(a_i)_{/T} \circ \mathrm{pr}_1 = (a_j)_{/T} \circ \mathrm{pr}_2$ as morphisms $(Y_i \times_Y Y_j)_{/T} \to X'_{/T}$ by the agreement with α (this uses that completion $_{/T}$ commutes with fibre products). By the injectivity already proven this shows that $a_i \circ \mathrm{pr}_1 = a_j \circ \mathrm{pr}_2$ as morphisms $Y_i \times_Y Y_j \to X'$. Since X' is an fppf sheaf this means that the collection of morphisms a_i descends to a morphism $a : Y \to X'$. We have $\alpha = a_{/T}$ because $\{(a_i)_{/T} : (Y_i)_{/T} \to X'_{/T}\}$ is an étale covering.

By the result of the previous paragraph, to prove existence, we may assume that Y is affine and that $g : Y \to X$ factors as $g_1 : Y \to X_1$ and an étale morphism $X_1 \to X$ with X_1 affine. Then we can consider $T_1 \subset |X_1|$ the inverse image of T and we can set $X'_1 = X' \times_X X_1$ with projection $f_1 : X'_1 \to X_1$ and

$$\alpha_1 = (\alpha, (g_1)_{/T_1}) : Y_{/T_1} = Y_{/T} \longrightarrow X'_{/T} \times_{X_{/T}} (X_1)_{/T_1} = (X'_1)_{/T_1}$$

We conclude that it suffices to prove the existence for α_1 over X_1, in other words, we may replace $X, T, X', Y, f, g, \alpha$ by $X_1, T_1, X'_1, Y, g_1, \alpha_1$. This reduces us to the case described in the next paragraph.

Assume Y and X are affine. Recall that $(Y_{/T})_{red}$ is an affine scheme (isomorphic to the reduced induced scheme structure on $g^{-1}T \subset Y$, see Formal Spaces, Lemma 0GB9). Hence $\alpha_{red} : (Y_{/T})_{red} \to (X'_{/T})_{red}$ has quasi-compact image E in $f^{-1}T$ (this is the underlying topological space of $(X'_{/T})_{red}$ by the same lemma as above).

Thus we can find an affine scheme V and an étale morpism $h : V \to X'$ such that the image of h contains E. Choose a solid cartesian diagram

$$\begin{array}{ccccc} Y'_{/T} & \dashrightarrow & W & \longrightarrow & V_{/T} \\ & \searrow & \downarrow & & \downarrow{\scriptstyle h_{/T}} \\ & & Y_{/T} & \xrightarrow{\alpha} & X'_{/T} \end{array}$$

By construction, the morphism $W \to Y_{/T}$ is representable by algebraic spaces, étale, and surjective (surjectivity can be seen by looking at the reductions, see Formal Spaces, Lemma 0GB8). By Lemma 0AR4 we can write $W = Y'_{/T}$ for $Y' \to Y$ étale and Y' affine. This gives the dotted arrows in the diagram. Since $W \to Y_{/T}$ is surjective, we see that the image of $Y' \to Y$ contains $g^{-1}T$. Hence $\{Y' \to Y, Y \setminus g^{-1}T \to Y\}$ is an étale covering. As f is an isomorphism over U we have a (unique) morphism $Y \setminus g^{-1}T \to X'$ over X agreeing with α on completions (as the completion of $Y \setminus g^{-1}T$ is empty). Thus it suffices to prove the existence for Y' which reduces us to the case studied in the next paragraph.

By the result of the previous paragraph, we may assume that Y is affine and that α factors as $Y_{/T} \to V_{/T} \to X'_{/T}$ where V is an affine scheme étale over X'. We may still replace Y by the members of an affine étale covering. By Lemma 0AR3 we may find an étale morphism $b : Y' \to Y$ of affine schemes which induces an isomorphism $b_{/T} : Y'_{/T} \to Y_{/T}$ and a morphism $c : Y' \to V$ such that $c_{/T} \circ b_{/T}^{-1}$ is the given morphism $Y_{/T} \to V_{/T}$. Setting $a' : Y' \to X'$ equal to the composition of c and $V \to X'$ we find that $a'_{/T} = \alpha \circ b_{/T}$, in other words, we have existence for Y' and $\alpha \circ b_{/T}$. Then we are done by replacing considering once more the étale covering $\{Y' \to Y, Y \setminus g^{-1}T \to Y\}$. □